Let T be the waiting time until the first text, and let $N(t)$ be the number of texts in the first t hours

$$N(t) \sim \text{Poisson}(\lambda t)$$

$$T > t \iff N(t) = 0$$

$$P(T > t) = P(N(t) = 0) = e^{-\lambda t}$$

$$p = P(1 < T < 2) = P(T > 1) - P(T > 2) = e^{-\lambda} - e^{-2\lambda}$$

Let $x = e^{\lambda}$

$$px^2 - x + 1 = 0$$

$$x = \frac{1 \pm \sqrt{1 - 4p}}{2p}$$

$\because 0 < p < \frac{1}{4}$

$$1 - 4p > 0$$

so there are two distinct real roots.

$$\frac{1 + \sqrt{1 - 4p}}{2p} > 1, \qquad \frac{1 - \sqrt{1 - 4p}}{2p} = \frac{2}{1 + \sqrt{1 - 4p}} > 1$$

$$\therefore x > 1$$

$$\therefore \lambda = \ln x > 0$$

There are two positive values of λ

---

e^{λ_1} and e^{λ_2} are roots of $px^2 - x + 1 = 0$

$$e^{\lambda_1} e^{\lambda_2} = \frac{1}{p} \implies \lambda_1 + \lambda_2 = \ln \frac{1}{p}$$

The two phones together have rate of $\lambda_1 + \lambda_2$

$$P(1 < T < 2) = e^{-(\lambda_1 + \lambda_2)} - e^{-2(\lambda_1 + \lambda_2)}$$

$$\because e^{\lambda_1 + \lambda_2} = \frac{1}{p}$$

$$\therefore \boxed{P(1 < T < 2) = p - p^2}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: June 25, 2026